Methyl Ethyl Ketone Economic Analysis

CHE 396 - Senior Design I

Homework No. 4

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Table of Contents

Project Charter

Title
Methyl Ethyl Ketone Economic Analysis
Goal
To analyze the cost of building and operating a Methyl Ethyl Ketone plant and determine the feasibility of the project based on the cost analysis spanning over the life of the plant.
In Scope
▪ Capital Cost ▪ Cost of Equipment ▪ ISBL, OSBL, Contingency, FCOP, VCOP ▪ Revenues, Gross Profit, ROI, Simple Payback Period ▪ NVP using MACRS depreciation ▪ DCFROR ▪ Sensitivity Analysis
Out of Scope
▪ Government Regulations ▪ Exact IRS tax applications ▪ Vendor Price Estimations ▪ Cost of Land and Legal Paperwork ▪ Side Products ▪ Economics of other possible utilities configurations
Deliverables
▪ Project Charter ▪ Design Plan ▪ Process Description ▪ Estimate of Capital Cost ▪ Estimate of Operations ▪ Estimates of Equipment ▪ Estimates of Revenues and Gross Profits ▪ Simple Payback Period and ROI ▪ NPV and DCFROR ▪ Sensitivity Analysis
Team Members
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Design Plan

Table 1: Design Plan

Methyl Ethyl Ketone Economic Analysis				
Task	Assignee	Start Date	End Date	Duration (Days)
First Meeting - Set Goals	ALL	11-Nov	11-Nov	1
Preliminary Research	ALL	11-Nov	12-Nov	2
Second Meeting - Divide Work	ALL	12-Nov	12-Nov	1
Equipment Hand Calculations	Emir & Evelyn	12-Nov	13-Nov	2
Aspen Analyzer Costs	Emir & Evelyn	13-Nov	13-Nov	1
Total Capital Cost	Emir & Evelyn	14-Nov	14-Nov	1
Variable Cost of Production (VCOP)	Evelyn & Sinan	14-Nov	14-Nov	1
Fixed Cost of Production (FCOP)	Sinan & Tariq	14-Nov	14-Nov	1
Revenue & Gross Profit	Emir & Evelyn	14-Nov	14-Nov	1
Third Meeting - Planning Execution	ALL	15-Nov	15-Nov	1
Simple Pay-Back Period & ROI	Emir & Evelyn	15-Nov	15-Nov	1
Net Present Value (NPV)	Emir & Evelyn & Tariq	16-Nov	16-Nov	1
Discounted Cash Flow Rate of Return (DCFROR)	Emir & Evelyn	17-Nov	17-Nov	1
Sensitivity Analysis	Emir & Evelyn & Tariq	17-Nov	17-Nov	1
Actual Report Writing	Emir & Evelyn	16-Nov	17-Nov	2
Editing and Formatting	Emir & Evelyn	17-Nov	Nov-19	1

Process Description

The production of Methyl Ethyl Ketone (MEK) is to be analyzed economically in order to determine the feasibility of the plant. A preliminary design was made for the plant to produce 10,000 tonnes per year. The equipment and their requirement are listed in Table 2**.** The plant has 8,000 hours per year as the operation rate and is to be built on a brown field with adequate infrastructure to provide the ancillary requirements of the new plant. Therefore, there is not an off-site investment requirement. The required raw materials and utilities are listed in Table 3. The construction of the plant will take two years in which the Capital will be 50% paid at the end of the first year and then the other 50% will be paid by the end of the second year. The production will be 50% in the first year of completion then it will be assumed that the rest of the 15-year plant life span will have full production.

- **Equipment Descriptions**

Table 2: Plant Equipment Specifications

	Equipment	Material	QTY	Description
1	Butanol Vaporizer	Carbon Steel	1	shell and tube heat exchanger, kettle type, heat transfer area 15m^2 , design pressure 5 bar, material carbon steel
2	Reactor Feed Heaters	Stainless Steel	2	shell and tube, fixed head, heat transfer area 25 m^2 , design pressure 5 bar, material stainless steel
3	Reactors	Stainless Steel	3	shell and tube construction, fixed tube sheets, heat transfer area 50 m^2, design pressure 5 bar, material stainless steel
4	Condenser	Stainless Steel	1	shell and tube heat exchanger, fixed tube sheets, heat transfer area 25 m^2, design pressure 2 bar, material stainless steel
5	Absorption Column	Carbon Steel	1	packed column, diameter 0.5 m, height 6.0 m, packing height 4.5 m, packing 25 mm ceramic saddles, design pressure 2 bar, material carbon steel
6	Extraction Column	Carbon Steel	1	packed column, diameter 0.5 m, height 4 m, packed height 3 m, packing 25 mm stainless steel pall rings, design pressure 2 bar, material carbon steel
7	Solvent Recovery Column	Carbon Steel	1	plate column, diameter 0.6 m, height 6 m, 10 stainless steel sieve plates, design pressure 2 bar, column material carbon steel
8	Recovery Column Reboiler	Carbon Steel	1	thermosyphon, shell and tube, fixed tube sheets, heat transfer area 4 m^2, design pressure 2 bar, material carbon steel
9	Recovery Column Condenser	Carbon Steel	1	double-pipe, heat transfer area 1.5 m^2, design pressure 2 bar, material carbon steel
10	Solvent Cooler	Stainless Steel	1	double pipe exchanger, heat transfer area 2 m^2, material stainless steel
11	Product Purification Column	Stainless Steel	1	plate column, diameter 1 m^2, height 20 m, 15 sieve plates, design pressure 2 bar, material stainless steel
12	Product Column Reboiler	Stainless Steel	1	kettle type, heat transfer area 4 m^2, design pressure 2 bar, material stainless steel
13	Product Column Condenser	Stainless Steel	1	shell and tube, floating head, heat transfer area 15 m^2, design pressure 2 bar, material stainless steel.
14	Feed Compressor	-	1	centrifugal, rating 750 kW
15	Butanol Storage Tank	Carbon Steel	1	cone roof, capacity 400 m^3, material carbon steel.
16	Solvent Storage Tank	Carbon Steel	1	horizontal, diameter 1.5 m, length 5 m, material carbon steel
17	Product Storage Tank	Carbon Steel	1	cone roof, capacity 400 m^3 , material carbon steel.

- **Utilities and Raw Materials**

Table 3: Raw Materials and Utilities Specifications

Raw Materials	
2-Butanol	1.045kg per kg of MEK
Solvent (trichloroethane)	Make-up 7,000 kg per year
Utilities	
Fuel Oil	3,000 metric tons per year, heating value 45GJ/metric ton
Cooling Water	120 metric tons per hour
Steam (low pressure)	1.2 metric tons per hour
Electrical Power	1MW

Capital Cost Estimate

In order to estimate the cost of the MEK plant, the Order of Magnitude Method was utilized by taking the capacity of an old established plant and its cost data and relating it to the new project's capacity. The equation below was implemented as shown with the data of a 1967 Methyl Ethyl Ketone plant that produced 35,000 tonnes per year [4]with an investment cost of $3,750,000. The cost from 1967 required updating to 2019 dollars by applying the respective year's indices to obtain the cost of $22,142,071.56 after calculating it with equation below.

$$C_{2019} = C_{1967} * \left(\frac{Cost\ Index_{2019}}{Cost\ Index_{196}}\right)$$

$$C_{201} = \$3{,}750{,}000 * \frac{647.7294}{109.7}$$

$$C_{2019} = \$22{,}142{,}071.56$$

Then the exponential estimation is shown below using the sixth-tenth rule scaling (n=0.6) since the plant Capital Cost can be obtained from the sum of the equipment to determine the estimate given the 10,000 metric tons per year production.

$$C_2 = C_1 * \left(\frac{S_2}{S_1}\right)^n$$

$$C_2 = \$22{,}142{,}071.56 * \left(\frac{10}{35}\right)^{0.6}$$

$$\boxed{C_2 = \$10{,}441{,}849.36}$$

After all the variables are obtained, the equation generates a Capital Cost of $10,441,849.36 for an MEK plant producing 10,000 metric tons per year.

Equipment Cost Calculations

Hand Calculations

One of the major components of analyzing the cost of the project it the equipment cost per unit as well as their installation cost. An estimation was completed using the cost curves which use the following equation from Towler's Textbook [6]:

$$C_e = a + bS^n$$

Table 7.2 "Purchase Equipment Cost for Common Plant Equipment" [6] was the source with all parts of the above equation. Variables a and b are constants; S is the size parameter with defined units, and n is the exponent for a particular equipment unit. It must be noted that by using the table some assumptions were made based on the provided equipment description to fit the available equipment's listed in the table. Also, the same S was not used for all the units, they all varied by their shape and size and some of the S's were out of the lower or upper bounds stated in the book but were still implemented in calculating the cost. Therefore, these calculations were taken as preliminary estimates which then needed to be multiplied by the installation factor based on the equipment type. The installation factors were gathered from Table 7.4 in Towler's Book which is included in Appendix

Sample Calculation for Equipment 11: Product Purification Column

The column is categorized with the following variables:

Table 4: Product Purification Specifications based on Cost Curves

Equipment Description	a	b	S(Shell Mass kg)	n
vertical 304 ss	17400	79	566.8986294	0.85

The S which in this equipment means shell mass was calculated, the first part needed was the thickness which is the following using Equation 14.13 from the textbook [6]:

$$thickness = \frac{P_i D_i}{(2SE - 1.2P_i)}$$

The design pressure should be 10% above the operating pressure which makes that 2.2bar rather than 2bar. Then the equation was solved like the following:

$$thickness = \frac{2.2\text{bar} * 10^5 * 1m}{(2 * 89 * 10^6 * 1.0) - (1.2 * 2.2bar * 10^5)}$$

$$= 0.001124m$$

The shell mass was then be found by:

$$Shell\ mass = \pi D_c L_c t_w \rho$$

The density of stainless steel is 8,030kg/m^3, the length is given as 20m which then it was calculated as the following:

$$Shell\ mass = \ \pi * 1m * 20m * 0.00124m * \frac{8{,}030\text{kg}}{\text{m}^3}$$

$$= 566.899\ kg$$

Using the shell mass, the cost of the column can was calculated using the following:

$$C_{vessel\ ss} = a + bS^n$$

$$C_{vessel\ ss} = 17{,}400 + 79 * 566.899^{0.85}$$
$$= \$34{,}702.41$$

The next step was to add the cost of the trays which also use the same cost curve but have different variables and even though only 15 are needed they are sold in packs of 30 therefore the cost was multiplied by the quantity.

Table 5: SS Sieve Tray variables for Cost Estimation

Equipment Description	a	b	S(Diameter m)	n
Ss Sieve Trays (sold in 30 per pack)	130	440	1	1.8

$$C_{ss\ sie\quad tray} = a + bS^n$$

$$C_{ss\ sieve\ trays} = (130 + 440 * 1^{1.8}) * 30trays$$
$$= \$17{,}100$$

Having both the column and the trays cost they were added to obtain the overall cost:

$$C_{total} = C_{vessel\ ss} + C_{ss\ sieve\ tra}$$
$$C_{total} = \$34{,}702.41 + \$17{,}100$$
$$= \$51{,}802.41$$

This total was multiplied by the installation factor of 4 that applies to distillation columns according to Table 7.4 in Towler's Textbook:

$$Total\ Cost\ with\ Installation\ Factor = 4 * \$51{,}802.41$$
$$= \$207{,}209.65$$

This amount was then converted to 2019 dollars by utilizing indices:

$$C_{2019} = C_{2010} * \left(\frac{Cost\ Index_{2019}}{Cost\ Index_{2010}}\right)$$

$$C_{2019} = \$207{,}209.65 * \frac{647.7294}{550.8}$$

$$= \$243{,}674.26$$

Each equipment followed the same general procedure as shown with the column above by utilizing the cost curves and installation factor multiplication which then the cost was adjusted to 2019 dollars from 2010. Some vary with the material. Others with the number of units and then some with the addition of packing. The total hand calculations costs are summarized in Table 6 below.

Table 6: Equipment with Installation Costs

Equipment	Hand Calculations Cost $2019	Installation Factor	Equipment + Installation Cost $2019
Butanol Vaporizer	$38,571.14	3.5	$134,998.99
Reactor Feed Heaters	$71,899.22	3.5	$251,647.26
Reactors	$139,895.21	3.5	$489,633.23
Condenser	$41,548.99	3.5	$145,421.46
Absorption Column	$18,729.93	4	$74,919.70
Extraction Column	$25,876.59	4	$103,506.35
Solvent Recovery Column	$25,711.02	4	$102,844.10
Recovery Column Reboiler	$36,408.98	3.5	$127,431.44
Recovery Column Condenser	$2,290.80	3.5	$8,017.79
Solvent Cooler	$2,306.43	3.5	$8,072.51
Product Purification Column	$60,918.57	4	$243,674.26
Product Column Reboiler	$35,741.40	3.5	$125,094.90
Product Column Condenser	$39,753.64	3.5	$139,137.74
Feed Compressor	$1,930,791.92	2.5	$4,826,979.80
Butanol Storage Tank	$131,548.02	4	$526,192.07
Solvent Storage Tank	$16,891.03	4	$67,564.12
Product Storage Tank	$131,548.02	4	$526,192.07
		TOTAL:	$7,901,327.79

▪ ASPEN Calculations

In order to compare the equipment hand calculation costs from using the equations in [6], the equipment costs using *Aspen Process Economic Analyzer V9* was used to see how accurate the equipment costs were. When using *Aspen*, the software allows the user to add a lot more details for the equipment in order to receive the best cost estimate. Although there is a lot of available parameters to specify for equipment cots, only a select amount of information was given such as the heat transfer areas, column heights, pressures, and materials. With all the empty slots of data, *Aspen* assigns default values to be able to complete the cost estimation in which it won't be a completely perfect estimation.

In calculating the cost estimate for the Purification Column, *Aspen* asks for several parameters before it can run the estimation. For this specific column, *Aspen* asks to enter the vessel diameter and the number of trays. The following table displays the specific cells in red that needs to be accounted for the purification column.

Table 7: Aspen Economic Analyze Data Input

Purification Column - Trayed tower

Name	Units	Item 1
Item Reference Number		3
Remarks 1		
Remarks 2		
Item description		Purification Column
User tag number		
Structure tag		
Component WBS		
Quoted cost per item	USD	
Currency unit for matl cost		
Source of quote		
Number of identical items		1
Installation option		
Code of account		
Icarus/User COA option		
Tray type		VALVE
Application		DISTIL
Shell material		
Vessel diameter	M	?
Vessel tangent to tangent height	M	
Design gauge pressure	KPAG	100
Vacuum design gauge pressure	KPAG	
Design temperature	DEG C	
Operating temperature	DEG C	
Tray material		A285C
Number of trays		?

Fortunately, there is additional data that can be inputted such as tray material, tray type, shell material, vessel tangent to tangent height, and design pressure in which a more accurate estimation can be acquired. After entering the remaining data, the equipment cost can then be calculated. The table below indicates the cost summary from *Aspen.*

Table 8: Economic Analyzer – Summary Costs

Summary Costs

Item	Material(USD)	Manpower(USD)	Manhours
Equipment&Setting	146000.	1762.	55
Piping	67220.	28606.	917
Civil	2066.	2657.	103
Structural Steel	13086.	2506.	85
Instrumentation	68343.	24578.	766
Electrical	2739.	1570.	51
Insulation	16470.	14475.	609
Paint	349.	649.	28
Subtotal	**316273**	**76803**	**2614**

The value that is comparable to the hand calculation result is the "Equipment & Setting" cell, at a total cost estimate of $146,000. Similar processes were done for the rest of the equipment that is necessary to run the MEK plant. The specific version of *Aspen Economic Analyzer* used provides the cost estimates in year 2015 dollars, so the appropriate conversions were made to convert the cost into 2019 dollars like that was done in the capital cost section. The corresponding vales for the cost index are in the appendix at the end Table 1A.

- ## Differences Between the Two Methods

When looking at the differences between the hand calculations and *Aspen* estimates, some equipment values are very similar while some others are very different. In conclusion, the *Aspen* results do have a slightly higher total cost of equipment by 11.74%. There were more details that can be included in the *Aspen* software in which is why this result occurs. The following table displays each equipment cost that were estimated from hand calculations and *Aspen* with their corresponding percent differences.

Table 9: Equipment Cost Estimates

	Equipment	Quantity	ASPEN COST $2019	HAND CALCULATIONS COST $2019	Percent Difference
1	Butanol Vaporizer	1	$37,164.17	$38,571.14	5.58
2	Reactor Feed Heaters	2	$62,120.60	$71,899.22	16.45
3	Reactors	3	$72,241.37	$139,895.21	65.46
4	Condenser	1	$19,659.89	$41,548.99	73.15
5	Absorption Column	1	$30,943.97	$18,729.93	47.42
6	Extraction Column	1	$25,011.10	$25,876.59	5.27
7	Solvent Recovery Column	1	$46,648.62	$25,711.02	56.15
8	Recovery Column Reboiler	1	$10,818.76	$36,408.98	109.68
9	Recovery Column Condenser	1	$3,257.26	$2,290.80	33.03
10	Solvent Cooler	1	$9,073.80	$2,306.43	117.72
11	Product Purification Column	1	$156,232.15	$60,918.57	86.27
12	Product Column Reboiler	1	$17,100.61	$35,741.40	72.18
13	Product Column Condenser	1	$18,263.92	$39,753.64	75.69
14	Feed Compressor	1	$2,245,978.70	$1,930,791.92	13.23
15	Butanol Storage Tank	1	$156,115.81	$131,548.02	15.22
16	Solvent Storage Tank	1	$26,872.39	$16,891.03	43.84
17	Product Storage Tank	1	$156,115.81	$131,548.02	15.22
	TOTAL		$3,093,618.93	$2,750,430.89	11.74

Total Capital Cost

▪ Inside Battery Limits (ISBL)

The ISBL cost refers to the cost it takes to run the plant "inside the fences" [6]. The main component in ISBL includes all the process equipment necessary to run the plant. Some other components that are including in ISBL are piping, buildings, and installation. For the capital cost calculations, equipment estimates from the hand calculations will be used as assumptions from [6] will be applied. In order to consider the other components of ISBL, table X contains all the installation factors for each equipment that will be applied to gain a more accurate value for ISBL.

▪ Outside Battery Limits (OSBL)

The OSBL cost attributes to any expansions from the existing site that can include new buildings, boilers, offices and labs. OSBL cost is normally 20-50% of the ISB [6]. Due to the MEK plant being built on an existing site with adequate infrastructure, there will not be any offsite investments which will result in an even lower percentage for OSBL. Thus, a fair assumption that can be made is that the OSBL is 15% of the ISBL.

▪ Contingency

The Contingency involves any variations from cost estimates, which result from labor disputes to currency fluctuations. Cost estimates are usually uncertain and are not exactly known until the specific item is installed completely. Therefore, contingency gives some range in price in case any of the disturbances mention occur. For the MEK plant, this technology is well known as other MEK plants do exist [9], in which the contingency assumption will be 10% the ISBL.

▪ Engineering & Construction

The Engineering costs include any detailed designed projects that are necessary to complete another project. This can include hiring engineers externally so they can carry out a specific part of a project that is essential in continuing that project in the future. These costs can include contractor's profit, project management, and plant layouts. The engineering and construction cost are about 10-30% of the ISBL depending on how large the projects may be. Thus, a safe assumption will be to have the engineering costs to be 20% of the ISBL.

The table below shows the analysis of each component with its corresponding total percentage, as well as the final total capital cost of the MEK plant.

Table 10: Breakdown of Capital Cost

Components of Capital Cost	Details	Cost
ISBL	Equipment + Installation	$7,901,327.79
OSBL	15% of ISBL	$1,185,199.17
Contingency	10% of ISBL	$790,132.78
Engineering & Construction	20% of ISBL & OSBL	$1,580,265.56
Working Capital	15% of ISBL	$1,185,199.17
Total Capital Cost		$12,642,124.46

Variable Cost of Production (VCOP)

Raw Materials & Utilities

The VCOP is represented by the cost of the plant's utilities and raw materials needed to produce the 10,000 metric tons of MEK in one year. The production requirement are displayed in Table 3 above**.** All the following components were converted to have units of USD/ kg to then be able to easily calculate the USD/yr except for the fuel oil which was left in USD/ GJ and electricity which is USD/MWh. Also depending on the sources' year of publication the prices were adjusted to fit the 2019 cost using respective indexes tabulated in Appendix Table 1A. The cost of 2-Butanol is around $1.05/kg [5] and the cost of the trichloroethane solvent was $2.36/kg [5] which are with their increase of about 4% each. The utilities cost are heavily reflective of the fuel oil cost since that is the most expensive out of the four with a cost of $5.69/GJ[7]. The electricity is $74.11/MWh [3], the cost of low-pressure steam is $10.44/kg [6] and the cost of water is the cheapest of all four with a cost of $0.000338/kg [8]. The cost per unit is then calculated to fit the yearly need of each one and the overall cost for the whole plant is then the addition of all the components which equal $12,686,996.80 as seen in Table 11 to produce the 10, 000 Metric Tons of MEK each year, this amount is determined to be the VCOP. The estimated USD/kg of MEK is $2[6] which in a year would amount to $20,000,000.00.

Table 11: Variable Costs of Production for MEK

Description	Quantity	Cost (USD/kg)	Total (USD/yr)
Product			
MEK(kg)	10000000	$2.00	$20,000,000.00
Raw Materials			
2-Butanol (kg/kg of MEK)	1.045	$1.05	$10,972,500.00
Solvent (trichloroethane) (kg/yr)	7000	$2.36	$16,490.80
Utilities			
Fuel Oil (MMTA)	3000	$5.69/GJ	$768,150.00
Fuel Oil Heating Value ($/GJ)	45		
Cooling Water (tonne/hr)	120	$0.000338	$324,480.00
Steam (low pressure) (Metric Ton/hr)	1.2	$10.44	$12,528.00
Electrical Power (MW)	1	$74.11/MWh	$592,848.00
Total			$12,686,996.80

Fixed Cost of Production (FCOP)

There are multiple components that contribute to the fixed cost of production such as labor, maintenance, land, and insurance. These components all have their fair share of their own percentages that contribute to the FCOP. The reason that the FCOP is 'fixed' is because even with a change of production rate, this cost of production will remain the same. This MEK plant will operate at 8,000 hours per year, which concludes to about 333 days in one year. It is expected for a plant not to operate continuously throughout entire year due to shutdowns that are needed to improve or replace any means of production. Since a typical plant has about 4.8 operators per shift [6], it was assumed for this MEK plant to have 5 operators per shift position. An average wage for a plant operator was found to be about $18.18 per hour[10], in which this wage and the number of operators per shift was used to calculating the operating labor component for the FCOP. As for the supervision for those operators, an average of 25% of the operating labor cost covers the supervision's salary, and about 45% of the operators plus the supervision cost covers any direct salary overhead employees. [6].

The remaining components of the FCOP all depending on the ISBL and OSBL costs that were calculated in the total capital cost section. The following table shows the calculated FCOP with each operation's contribution. The percentages that each component has that correspond to the ISBL and OSBL cost were found in Towler's textbook [6].

Table 12: Fixed Cost of Production for MEK

Operations	Description	Cost
Operating Labor	5 Operators | $18.18/hr	$727,200.00
Supervision	25% Operating Labor	$181,800.00
Direct Salary Overhead	45% Operating Labor + Supervision	$409,050.00
Maintenance	5% ISBL	$395,066.39
Property Taxes & Insurances	2% ISBL + OSBL	$181,730.54
Rent of Land	1.5% ISBL + OSBL	$136,297.90
General Plant Overhead	65% Operating Labor + Maintenance	$729,473.15
Environmental Charges	1% ISBL + OSBL	$90,865.27
Royalties	7% Revenue	$1,400,000.00
Capital Charges	6% Working Capital	$71,111.95
Sales & Marketing	?	$0.00
TOTAL:		$4,322,595.21

Revenues and Profits

- **Revenue**

The production of Methyl Ethyl Ketone has a selling price of $2/kg as stated in the VCOP section above which means that the revenue earned from the plant is the following:

$$\frac{\$2}{kg} * \frac{10{,}000 metrictons}{year} * \frac{1{,}000kg}{1metric\ ton}$$

Revenue= $20,000,000/year

- **Gross Profit**

The gross profit for a chemical plant is roughly calculated to be the following.

$$Gross\ profit\ =\ product\ value - (1.2\ \times\ raw\ materials\ costs)$$

= $6,813,211.04

A more accurate number is from the average annual cash flow which is the Revenue minus the CCOP which is equal to

$2,990,407.99

- **Simple Pay-Back Period**

Simple payback period is the estimated time it will take an investment to be paid for dividing it by the average cash flow. This equation assumes that all the investment was made in the year zero and the profits start right away. We know this is not the realistic calculation since in the design process it was stated that the plan was going to take two years to construct and in year three the production will be 50%. Therefore, this equation is purely based on cash flow and taxed and depreciation are also disregarded with the annual income used rather than the cash flow. The following is the Equation 9.15 from the textbook [6]

$$simple\ payback\ time\ =\ \frac{total\ investment}{average\ annual\ cash\ flow}$$

$= 4.23 years$

- **Return on Investment (ROI)**

The return on investment is a simple economic performance measurement. The net annual profit is divided by the total investment and then multiplied by 100%. The net annual profit is the after tax annual operating income. The following are Equations 9.16 and 9.17 from the textbook [6]. Equation 9.17 is for the ROI over the whole project's life.

$$ROI\_a\ = \frac{net\ annualprofit}{total\ investment} *\ 100\%$$

=23.65%

$$ROI\_b = \frac{cumulative\ net\ profit}{plant\ life * initial\ investment} * \ 100\%$$

=1.58%

- ## Cash Cost of Production (CCOP)

  The CCOP is just the addition of the FCOP and VCOP as shown below in **TABLE x.**

Table 13: Summary of Production Costs

Cost of Production Spec	Cost
Variable	$12,686,996.80
Fixed	$4,322,595.21
Cash	$17,009,592.00

**All the Revenue and Profit sections are summarized in Table 14 to display the concise data needed for the proceeding sections.

Table 14: Revenues and Profits Summary

FCOP:	$4,322,595.21
VCOP:	$12,686,996.80
REVENUE:	$20,000,000.00
TOTAL INVESTMENT:	$12,642,124.46
CCOP:	$17,009,592.01
AVG ANNUAL CASH FLOW:	$2,990,407.99
Simple Payback Time:	4.23years
ROI_a:	23.65%
ROI_b:	1.58%
Gross Profit:	$6,813,211.04

Economic Evaluation

▪ Net Present Value (NPV)

The NPV is the sum of present cash flow values, that account for the time value of money in the future. Factors that contribute to the NPV include pre-tax cash flow, depreciation, and tax rate. Depreciation is the reduction of value and that is dependent on the specific recovery time. Depreciation values were obtained from Table 15. The following formula was used in calculating the NPV for 5- and 7-year recovery.

$$NPV = \sum_{i=1}^{t} \frac{CF_n}{(1+i)^n}$$

When calculating the NPV, certain specifications of the MEK plant were accounted for. One specification of the plant is that the capital cost will be paid in two equal increments at the end of the first and second year. Thus, this creates our pre-tax cash flow to be a negative value of half of the total capital cost for the first two years. Another specification that was account for was that once the two-year construction period ends, the plant will run at a 50% production rate. This leads to the plant's third year of pre-tax cash flow to be half of the average annual cash flow. As for the tax rate, it was assumed to be at 35% [6] and the discount rate for the MEK plant is 14%. The tables below present the calculations of the NPV for a 5- and 7-year recovery period over a 15-year span.

Table 15: NPV with 5-Year Recovery

				Interest Rate =		0.14		
Year	Pre-Tax CF (MM)	Depreciation	Taxable Income (MM)	Tax (MM)	CF (MM)	Fd	PV (MM)	NPV (MM)
0	-$6.32	0.00	-$6.32	$0.00	-$6.32	1.00	-$6.32	-$6.32
1	-$6.32	0.00	-$6.32	$0.00	-$6.32	0.88	-$5.54	-$11.87
2	$1.50	20.00	-$18.50	-$2.21	$3.71	0.77	$2.85	-$9.01
3	$2.99	32.00	-$29.01	-$6.48	$9.47	0.67	$6.39	-$2.62
4	$2.99	19.20	-$16.21	-$10.15	$13.14	0.59	$7.78	$5.16
5	$2.99	11.52	-$8.53	-$5.67	$8.66	0.52	$4.50	$9.66
6	$2.99	11.52	-$8.53	-$2.99	$5.98	0.46	$2.72	$12.38
7	$2.99	5.76	-$2.77	-$2.99	$5.98	0.40	$2.39	$14.77
8	$2.99	0.00	$2.99	-$0.97	$3.96	0.35	$1.39	$16.16
9	$2.99	0.00	$2.99	$1.05	$1.94	0.31	$0.60	$16.76
10	$2.99	0.00	$2.99	$1.05	$1.94	0.27	$0.52	$17.28
11	$2.99	0.00	$2.99	$1.05	$1.94	0.24	$0.46	$17.74
12	$2.99	0.00	$2.99	$1.05	$1.94	0.21	$0.40	$18.14
13	$2.99	0.00	$2.99	$1.05	$1.94	0.18	$0.35	$18.50
14	$2.99	0.00	$2.99	$1.05	$1.94	0.16	$0.31	$18.81
15	$2.99	0.00	$2.99	$1.05	$1.94	0.14	$0.27	$19.08

Table 16: NPV with 7-Year Recovery

				Interest Rate=	0.14			
Year	Pre-Tax CF (MM)	Depreciation	Taxable Income (MM)	Tax (MM)	CF (MM)	Fd	PV (MM)	NPV (MM)
0	$0.00	0.00	$0.00	$0.00	$0.00	1.00	$0.00	$0.00
1	$0.00	0.00	$0.00	$0.00	$0.00	0.88	$0.00	$0.00
2	$0.00	14.29	-$14.29	$0.00	$0.00	0.77	$0.00	$0.00
3	$0.00	24.49	-$24.49	-$5.00	$5.00	0.67	$3.38	$3.38
4	$0.00	17.49	-$17.49	-$8.57	$8.57	0.59	$5.08	$8.45
5	$0.00	12.49	-$12.49	-$6.12	$6.12	0.52	$3.18	$11.63
6	$0.00	8.93	-$8.93	-$4.37	$4.37	0.46	$1.99	$13.62
7	$0.00	8.92	-$8.92	-$3.13	$3.13	0.40	$1.25	$14.87
8	$0.00	8.93	-$8.93	-$3.12	$3.12	0.35	$1.09	$15.97
9	$0.00	4.46	-$4.46	-$3.13	$3.13	0.31	$0.96	$16.93
10	$0.00	0.00	$0.00	-$1.56	$1.56	0.27	$0.42	$17.35
11	$0.00	0.00	$0.00	$0.00	$0.00	0.24	$0.00	$17.35
12	$0.00	0.00	$0.00	$0.00	$0.00	0.21	$0.00	$17.35
13	$0.00	0.00	$0.00	$0.00	$0.00	0.18	$0.00	$17.35
14	$0.00	2.40	-$2.40	$0.00	$0.00	0.16	$0.00	$17.35
15	$0.00	0.00	$0.00	-$0.84	$0.84	0.14	$0.12	$17.47

- ## Results

Looking at the last row for each NPV value, it was estimated for the 5-year recovery period the NPV was about 19.1 million and for the 7-year recovery was about 17.4 million. A higher NPV value indicates that the projected earnings will exceed the anticipated cost in present dollars [11],, which leads to a more profitable project. Therefore, by looking at the calculated NPV values for each recovery period, it would be a superior choice to invest in the 5-year recovery period option.

- **Discounted Cash Flowrate of Return (DCFROR)**

The DCFROR is also known as the internal rate of return IRR in which the interest rate at the end of a project's life is set to a zero NVP. The higher the DCFROR the more profitable the project is projected to be, this is proven to be more useful than NVP when having to compare projects with varying sizes. Since DCFROR is hand in hand with interest rate it helps in comparing projects bases on other investments. The DCFROR ultimately measure the max interest rate that the project can pay and by the end of the project's life still break even.

$$\sum_{n=1}^{n=t} \frac{CF_n}{(1+i')^n} = 0$$

The DCFROR is found by implementing goal seek in excel from the respective 5 and 7 year term NVP. Goal seek set the last NVP value to zero by varying the interest. Tables 17 and 18 show the data below.

Table 17: DCFROR with 5-year Recovery

					Interest Rate=	0.450258		
Year	**Pre-Tax CF (MM)**	**Depreciation**	**Taxable Income (MM)**	**Tax (MM)**	**CF (MM)**	**Fd**	**PV (MM)**	**NPV (MM)**
0	-$6.32	0.00	-$6.32	$0.00	-$6.32	1.00	-$6.32	-$6.32
1	-$6.32	0.00	-$6.32	$0.00	-$6.32	0.69	-$4.36	-$10.68
2	$1.50	20.00	-$18.50	-$2.21	$3.71	0.48	$1.76	-$8.92
3	$2.99	32.00	-$29.01	-$6.48	$9.47	0.33	$3.10	-$5.81
4	$2.99	19.20	-$16.21	-$10.15	$13.14	0.23	$2.97	-$2.84
5	$2.99	11.52	-$8.53	-$5.67	$8.66	0.16	$1.35	-$1.49
6	$2.99	11.52	-$8.53	-$2.99	$5.98	0.11	$0.64	-$0.85
7	$2.99	5.76	-$2.77	-$2.99	$5.98	0.07	$0.44	-$0.41
8	$2.99	0.00	$2.99	-$0.97	$3.96	0.05	$0.20	-$0.20
9	$2.99	0.00	$2.99	$1.05	$1.94	0.04	$0.07	-$0.14
10	$2.99	0.00	$2.99	$1.05	$1.94	0.02	$0.05	-$0.09
11	$2.99	0.00	$2.99	$1.05	$1.94	0.02	$0.03	-$0.06
12	$2.99	0.00	$2.99	$1.05	$1.94	0.01	$0.02	-$0.03
13	$2.99	0.00	$2.99	$1.05	$1.94	0.01	$0.02	-$0.02
14	$2.99	0.00	$2.99	$1.05	$1.94	0.01	$0.01	-$0.01
15	$2.99	0.00	$2.99	$1.05	$1.94	0.00	$0.01	$0.00

Table 18: DCFROR with 7-year Recovery

					Interest Rate=	0.406077		
Year	**Pre-Tax CF (MM)**	**Depreciation**	**Taxable Income (MM)**	**Tax (MM)**	**CF (MM)**	**Fd (MM)**	**PV (MM)**	**NPV (MM)**
0	-$6.32	0.00	-$6.32	$0.00	-$6.32	1.00	-$6.32	-$6.32
1	-$6.32	0.00	-$6.32	$0.00	-$6.32	0.71	-$4.50	-$10.82
2	$1.50	14.29	-$12.79	-$2.21	$3.71	0.51	$1.88	-$8.94
3	$2.99	24.49	-$21.50	-$4.48	$7.47	0.36	$2.69	-$6.25
4	$2.99	17.49	-$14.50	-$7.52	$10.52	0.26	$2.69	-$3.56
5	$2.99	12.49	-$9.50	-$5.07	$8.07	0.18	$1.47	-$2.10
6	$2.99	8.93	-$5.94	-$3.32	$6.32	0.13	$0.82	-$1.28
7	$2.99	8.92	-$5.93	-$2.08	$5.07	0.09	$0.47	-$0.81
8	$2.99	8.93	-$5.94	-$2.08	$5.07	0.07	$0.33	-$0.48
9	$2.99	4.46	-$1.47	-$2.08	$5.07	0.05	$0.24	-$0.25
10	$2.99	0.00	$2.99	-$0.51	$3.50	0.03	$0.12	-$0.13
11	$2.99	0.00	$2.99	$1.05	$1.94	0.02	$0.05	-$0.08
12	$2.99	0.00	$2.99	$1.05	$1.94	0.02	$0.03	-$0.05
13	$2.99	0.00	$2.99	$1.05	$1.94	0.01	$0.02	-$0.03
14	$2.99	0.00	$2.99	$1.05	$1.94	0.01	$0.02	-$0.01
15	$2.99	0.00	$2.99	$1.05	$1.94	0.01	$0.01	$0.00

- ## MACRS Depreciation 5-year and 7-year Term

The equipment loses value after its purchased which in technical terms is known as depreciation. This benefits businesses since that means the taxes needed to be paid are less until the equipment or any asses gets to be 100% depreciated from what it cost when it was bought. The Modified Accelerated Cost Recovery System (MACRS) is a way in which the amount of years and amount of depreciation are varied to help save the most money. The table utilized for this project is taken from [6] and shown in Appendix.

- ## Results

In calculating the DCFROR for each recovery period, it was found the 5-year recovery period had a DCFROR of about 45%, while the 7-year recovery had about a 40% maximum interest rate. What this means is that since the 5-year recovery period has a higher value of maximum interest rate it is the more profitable option. This ultimately predicts that the project can pay up to 45% interest rate by the end of the project's life still break even rather than 40% from the 7-year recovery term.

Sensitivity Analysis

"The Sensitivity Analysis gives a qualitative indication of the relative sensitivity of NPV to major cost and process parameters." [6] The graph is plotted with a base of $19.08 million and then each component under inspection was analyzed by varying the percent up to +40% and -40% of the base line. The cost of utilizes, raw materials, and the profit (product cost) are all the varying parts since they are uncertain and changing as time progresses. The plot below, Figure 1, helps to inspect the inherent risk that comes from investing in the MEK plant. The 5-year depreciation model was the chosen one since it was discovered to be more profitable based on the NVP and DCFROR above. As seen below, the risk of the MEK plant relies heavily on the profit. A decrease of about 28% or more makes for an unprofitable plant investment while the inverse of that makes it twice as profitable from the base NVP. The utilities and the raw material variances are less critical than the profits. Also, it can be interpolated that if the material is varied 45% above the base the project is unprofitable. While the utilities would take much more varying to obtain a plant that makes no profit.

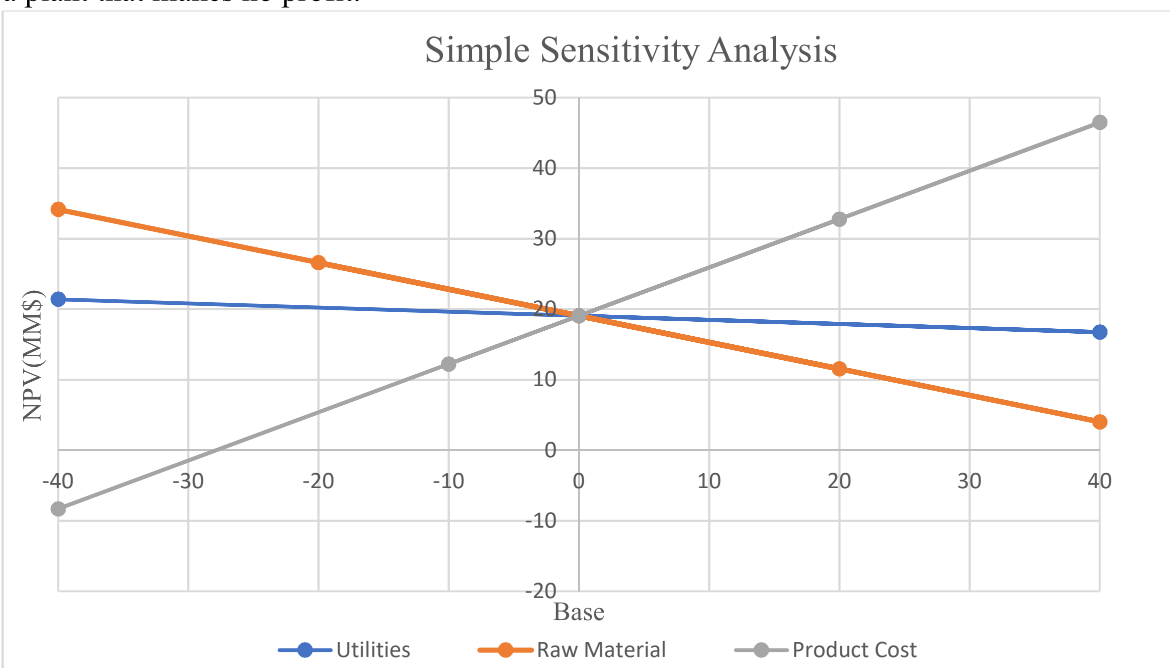

Figure 1: Graph for 5-year Recovery Term NVP

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APPENDIX

Table 1A: Index [6]

Year	Index
1967	109.7
2001	394
2010	550.8
2013	567.3
2015	556.8
2018	603.1
2019	647.7

[2]

$$C_A = C_B * \left(\frac{Cost\ Index_A}{Cost\ Index_B}\right)$$

Table 7.4 Installation Factors Proposed by Hand (1958)

Equipment Type	Installation Factor
Compressors	2.5
Distillation columns	4
Fired heaters	2
Heat exchangers	3.5
Instruments	4
Miscellaneous equipment	2.5
Pressure vessels	4
Pumps	4

[6]

Table 9.1 MACRS Depreciation Charges

Recovery Year	Depreciation Rate ($F_i = D_i/C_d$)		
	5-Year Recovery	**7-Year Recovery**	**15-Year Recovery**
1	20	14.29	5.00
2	32	24.49	9.50
3	19.2	17.49	8.55
4	11.52	12.49	7.70
5	11.52	8.93	6.93
6	5.76	8.92	6.23
7		8.93	5.90
8		4.46	5.90
9			5.91
10			5.90
11			5.91
12			5.90
13			5.91
14			5.90
15			5.91
16			2.95

[6]